

## DIGITAL OPTICAL ROTARY ENCODER



### Technical Details:

|                  |                      |                                                                                            |
|------------------|----------------------|--------------------------------------------------------------------------------------------|
| Item             |                      | Diameter 38mm shaft 6mm type Incremental rotary encoder                                    |
| Resolution (P/R) |                      | 100, 360, 400, 600, 1000 PPR                                                               |
| Input Phase      |                      | AB Phase                                                                                   |
| Output Phase     |                      | NPN Open Collector                                                                         |
|                  | Supply Voltage       | 5-24 VDC                                                                                   |
| Electrical       | Current Consumption  | Max , 40mA                                                                                 |
|                  | Response Frequency   | Max , 100 Khz                                                                              |
|                  | Allowable Revolution | Max , 3000 rev / min                                                                       |
|                  |                      |                                                                                            |
| Mechanical       | Starting torque      | Max , 20 gf.cm ( 0.002N.m )                                                                |
|                  | Rotor inertia        | Max , 15 g.cm <sup>2</sup> ( 1,5* 10 <sup>-6</sup> kg.m <sup>2</sup> )                     |
|                  | Shaft Loading        | Radial : Max 2kgf, Axial : Max .1kgf                                                       |
|                  | Mechanical Speed     | Max .5000 rev / min (*1)                                                                   |
| Environmental    | Ambient Temperature  | -10~ 70 ( at non- freezing status ) , Stronger :- 25 ~ 85                                  |
|                  | Ambient Humidity     | 35 ~85% RH , Stronger : 35~ 90 % RH                                                        |
|                  | Protection           | Ip52 (IEG Standard )                                                                       |
|                  | Vibration            | 1.5 mm amplitude at frequency of 10-55 Hz in each Of X,Y,Z direction for 2 hour.           |
|                  | Shock                | Max. 40G                                                                                   |
| Unit Weight      |                      | Approx :180g                                                                               |
| Cable            |                      | 2.0m (the cable length can be customized )                                                 |
| Approval         |                      | CE ROHS                                                                                    |
| (*1 )            |                      | Mechanical speed > Allowable revolution , Please take allowable speed as standard when use |

| Wire Colour | OC VP OP 3-CHANNEL | TTL / HTL 6- CHANNEL | PIN 9-POLE | EXPLANATION    |
|-------------|--------------------|----------------------|------------|----------------|
| RED         | VCC                | VCC                  | 1          | Supply Voltage |
| BLACK       | 0 V                | 0 V                  | 4          | Common Port    |
| GREEN       | A                  | A                    | 5          | Signal wire    |
| WHITE       | B                  | B                    | 3          | Signal wire    |
| YELLOW      | Z                  | Z                    | 8          | Signal wire    |
| BROWN       | —                  | -A                   | 6          | Signal wire    |
| GRAY        | —                  | -B                   | 7          | Signal wire    |
| ORANGE      | —                  | -Z                   | 2          | Signal wire    |
| SHIELD      | SHIELD             | SHIELD               | 9          |                |

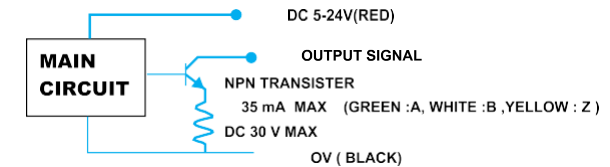
### Output:

AB two-phase quadrature output rectangular pulse, the circuit output is NPN open collector output type.

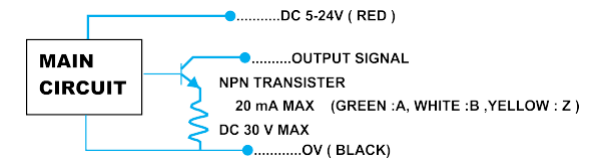
This type can be output with internal pull -up resistor available in Arduino, microcontroller s or PLC, such asAtmega, pic, 51 or microcontroller Mitsubi shi PLC.

If Internal Pullup is not available, then you need to Pull-up Output Channel A & B with resistors supplied with product. i.e. resister between Green and RED, White and RED wire.

#### OC NPN OPEN COLLECTOR



#### OC VOLTAGE OUTPUT



LD : LONG DRIVER OUTPUTS 5V  
LDH : LONG DRIVER OUTPUTS 5-24V

